

Nick Yaeger

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EDUCATION

B.S. in Electrical & Computer Engineering, Carnegie Mellon University

Expected 2028

GPA: 3.8

Relevant Coursework: Computer Architecture, Embedded Systems, Distributed Systems, Computer Vision.

EXPERIENCE

Instructor

Jan 2025 - Present

Student College, CMU Robotics Institute

Pittsburgh, PA

- Teach master's elective robotics course with enrollment spanning undergraduate, graduate levels.
- Give lectures, labs covering circuits, sensing, control, computer vision, actuators, path planning.
- Manage \$5K in funding from CMU Robotics Institute to provide low-cost robot kits to students.
- Expanded course to include capstone final project, oversaw 8-year record in student enrollment.

Embedded Systems Engineer

Aug 2024 - Present

Carnegie Mellon Racing

Pittsburgh, PA

- Design STM32-based PCB housing vehicle controls, cooling system, accumulator power sensing.
- Develop firmware in C and Python for autonomous driving, data acquisition, live telemetry.
- Implemented data acquisition pipeline with automotive-grade sensors, PCBs, CAN bus.
- Educate newer members on PCB design, firmware, embedded systems, engineering design cycle.

RESEARCH

Constrained Reinforcement Learning: Design and analyze various reinforcement learning algorithms under realistic hardware constraints and tradeoffs (memory, precision, latency).

Underwater Boat-Cleaning Robot: Develop autonomous robot capable of cleaning and inspecting boat hulls autonomously. Secured \$7.5K in grants.

PROJECTS

RISC-V Processor: Designed and tested RV32I-compliant CPU core with 7-stage pipeline, branch target buffer with two-bit counter, instruction and data caching. Achieved 289 MHz clock rate, 0.98 instructions per cycle. *SystemVerilog, Synopsys Verilog Compiler Simulator, Synopsys Design Compiler*

Chron Smartwatch: Build custom, open source smartwatch with heart rate monitor, oximeter, pedometer, Bluetooth, ML-based raise-to-wake, multi-week battery life. Currently prototyping. *Nordic nRF SoC, ZephyrRTOS, ARM, Bluetooth Low Energy, Altium Designer*

Real-Time Operating System: Implemented RTOS kernel in C/ARM assembly with context switching, thread management, rate-monotonic scheduling to support concurrent real-time tasks. Added admission control, execution-time/period enforcement, mutex synchronization with Highest Locker Protocol. *ARM, GDB, STM32*

LEADERSHIP

- **Senior Member, CMU Robotics Club:** Maintain club workshop and provide guidance to newer members. Mentor for CMU's Red Robot hackathon.
- **Eagle Scout, BSA:** Led team of 20+ volunteers to repaint and renovate parking lot for local high school, saving the school over \$1K and providing 140 volunteer hours.

SKILLS & INTERESTS

Technical Skills

C, Python, Go, Assembly, SystemVerilog, Vivado, Altium Designer, Linux, ROS2

Interests

Piano, Marching Band, Rock Climbing, Backpacking, Watching the Houston Astros